Using-cpis

Contents

R setup

```
suppressMessages({
  library(survival)
  library(quantreg)
  library(dplyr)
  library(knitr)
  library(randomForestSRC)
  library(ggplot2)
  library(tibble)
  library(cpis)
})
#> Warning: package 'survival' was built under R version 4.3.3
```

```
#' Performs conformal inference using various user-specified methods.
#' Supports both marginal and subgroup coverage guarantees
#'
#' @param dat_train Training data
#' @param dat_test Testing data
#' @param max.t.adjust If TRUE, limit to test times  max failed time in training data
#' @param alpha The miscoverage level (e.g., 0.1 corresponds to 90% coverage).
#' @param side Options are "two" for two-sided or "one" for one-sided.
#' @param split Numeric. The proportion of training data used to fit the working model.
#' The remaining (1 - split) portion is reserved for quantile regression.
#' @param subgroup Number of subgroups
#' @param methods A character vector specifying the conformal inference methods to apply.
#' Options include proposed CPI methods that ensure both marginal and subgroup
#'  coverage ("prop.weibull", "prop.cox", "prop.rf"),
#' and naive methods that ensure only marginal coverage but not subgroup coverage
#'  ("marginal.weibull", "marginal.cox", "marginal.rf").
#' The suffix (after the dot) indicates the working model used.
#'
#' @return
#'   - Coverage rates for marginal and subgroups (when side ="one")
#'   - Coverage rates, lower bound, upper bound and length (when side="two")
```

```
##### two-sided CPI-Weibull
main_fun(dat_train, dat_test,
                max.t.adjust=TRUE,
                alpha=0.10,side="two",
                split=0.5,
                subgroup=3,
                methods="prop.weibull")
#> $prop.weibull
#>         Method   Subgroup  Coverage        Lower     Upper    Length
```

```
#> 1 prop.weibull   Marginal 0.8983240 4.994804e-02 43.114277 43.064329
#> 2 prop.weibull Subgroup 1 0.9047619 4.611227e-03 64.583707 64.579095
#> 3 prop.weibull Subgroup 2 0.9189189 7.315141e-05 44.269979 44.269906
#> 4 prop.weibull Subgroup 3 0.8651685 1.804390e-03  8.305052  8.303247
```

```
##### two-sided CPI-Weibull, CPI-Cox and CPI-Random Forest
main_fun(dat_train, dat_test,
         max.t.adjust=TRUE,
         alpha=0.10,side="two",
         split=0.5,
         subgroup=3,
         methods=c("prop.weibull","prop.cox","prop.rf"))
#> $prop.weibull
#>         Method   Subgroup  Coverage        Lower     Upper    Length
#> 1 prop.weibull   Marginal 0.9002217 0.0199975526 31.035397 31.015400
#> 2 prop.weibull Subgroup 1 0.8571429 0.0025827568 17.122550 17.119967
#> 3 prop.weibull Subgroup 2 0.8181818 0.0000455379  7.541075  7.541029
#> 4 prop.weibull Subgroup 3 0.8851852 0.0011911629  9.286998  9.285806
#>
#> $prop.cox
#>     Method   Subgroup  Coverage        Lower     Upper    Length
#> 1 prop.cox   Marginal 0.8991131 1.838325e-02 32.136775 32.118391
#> 2 prop.cox Subgroup 1 0.8775510 2.194089e-03 22.972104 22.969910
#> 3 prop.cox Subgroup 2 0.8074866 3.810825e-05  8.902639  8.902601
#> 4 prop.cox Subgroup 3 0.8962963 2.875056e-03 13.679728 13.676853
#>
#> $prop.rf
#>    Method   Subgroup  Coverage     Lower     Upper    Length
#> 1 prop.rf   Marginal 0.8830377 0.3073173 10.794458 10.487141
#> 2 prop.rf Subgroup 1 0.8843537 0.0409356  6.745466  6.704531
#> 3 prop.rf Subgroup 2 0.8823529 0.1004736  9.593058  9.492585
#> 4 prop.rf Subgroup 3 0.8944444 0.1529719  7.604479  7.451507
```

```
##### two-sided Margianl-Weibull, Margianl-Cox and Margianl-Random Forest
main_fun(dat_train, dat_test,
         max.t.adjust=TRUE,
         alpha=0.10,side="two",
         split=0.5,
         subgroup=3,
         methods=c("marginal.weibull","marginal.cox","marginal.rf"))
#> $marginal.weibull
#>             Method   Subgroup  Coverage        Lower     Upper    Length
#> 1 marginal.weibull   Marginal 0.9230769 0.0057164046  82.88414  82.87842
#> 2 marginal.weibull Subgroup 1 0.8639456 0.0077187401 111.91670 111.90898
#> 3 marginal.weibull Subgroup 2 0.6935484 0.0008810242  12.77428  12.77340
#> 4 marginal.weibull Subgroup 3 0.8861940 0.0015430417  22.37310  22.37155
#>
#> $marginal.cox
#>         Method   Subgroup  Coverage       Lower    Upper   Length
#> 1 marginal.cox   Marginal 0.9286511 0.009067271 65.78960 65.78053
#> 2 marginal.cox Subgroup 1 0.8639456 0.011350859 70.05854 70.04719
#> 3 marginal.cox Subgroup 2 0.7204301 0.000605505 15.12689 15.12628
#> 4 marginal.cox Subgroup 3 0.9104478 0.001520669 36.20865 36.20713
#>
```

```
#> $marginal.rf
#>        Method   Subgroup  Coverage      Lower     Upper    Length
#> 1 marginal.rf   Marginal 0.8952062 0.25427951 11.094621 10.840342
#> 2 marginal.rf Subgroup 1 0.9047619 0.01966518 14.853030 14.833364
#> 3 marginal.rf Subgroup 2 0.8118280 0.13914129  7.832429  7.693288
#> 4 marginal.rf Subgroup 3 0.8917910 0.13850704  7.459740  7.321233
```

```
##### one-sided CPI-Weibull
main_fun(dat_train, dat_test,
         max.t.adjust=TRUE,
         alpha=0.05,side="one",
         split=0.5,
         subgroup=3,
         methods="prop.weibull")
#>               Marginal Subgroup 1 Subgroup 2 Subgroup 3
#> prop.weibull 0.9423503  0.9455782  0.9358289  0.9407407
```

```
##### one-sided CPI-Weibull, CPI-Cox and CPI-Random Forest
main_fun(dat_train, dat_test,
         max.t.adjust=TRUE,
         alpha=0.05,side="one",
         split=0.5,
         subgroup=3,
         methods=c("prop.weibull","prop.cox","prop.rf"))
#>               Marginal Subgroup 1 Subgroup 2 Subgroup 3
#> prop.weibull 0.9453735  0.9591837  0.9193548  0.9570896
#> prop.cox     0.9464883  0.9659864  0.9247312  0.9589552
#> prop.rf      0.9554069  0.9931973  0.9408602  0.9626866
```